

## Samuel M. Maione

[smaione1@jh.edu](mailto:smaione1@jh.edu) | <https://sammaione.github.io/>

### EDUCATION

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#### Binghamton University

*Thesis advisor:* Peter Gerhardstein

BSc in Integrative Neuroscience | Sociology (Minor)

*Binghamton, NY*

*May 2023*

### RESEARCH EXPERIENCE

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#### Laboratory Technician / Manager

Look, Infer, Understand (LIU) Lab

*Principal Investigator:* Shari Liu

*Johns Hopkins University*

*July 2023 - Present*

- Pre-registered and conducted novel research to investigate the relationship between the neural correlates of spatial working memory and intuitive physics using a series of fMRI analyses
- Mentored part-time research staff during their own fMRI analysis
- Created lab standard for fMRI analysis pipeline (based in Python, Jupyter, Bash)
- Created lab standard for looking time analysis using computer vision (based in Python, Jupyter, Bash)
- Created IRB protocols for fMRI, behavioral testing, and developmental studies
- Managed lab budget and reimbursement procedures for all members of the lab
- Organized lab events, inventory, project board, and lab manual
- Presented research to lab members, department members (Psychological and Brain Sciences Seminar), and outside lab members (Harvard Laboratory for Developmental Studies)

#### Research Assistant

Binghamton Baby Lab

*Principal Investigator:* Peter Gerhardstein

*Binghamton University*

*Jan 2022 - June 2023*

- Led partnership with Victoria Pelak (University of Colorado) and Peter Gerhardstein to create a novel survey to characterize the psychological and epidemiological profile of visual snow syndrome
- Completed an honors thesis and earned the Undergraduate Research Award and Steven W. Kovacs Memorial Award for my commitment to psychological research
- Ran in-person experiments with adults and children (aged 3 to 9 years)
- Programmed studies for EEG (using Matlab), eye tracking (using EPrime), and behavioral experiments (in PsychoPy)

### PUBLICATIONS

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**Maione, S. M., Pelak, V. S., & Gerhardstein, P. C.** Assessment of a Novel Patient Reported Outcome Measure for Visual Snow Syndrome: The Colorado Visual Snow Survey 2.0. *Frontiers in Neurology*, 16, 1664310. [[OSF](#)] [[Paper](#)]

**Maione, S. M., & Liu, S.** (in prep). Shared neural resources between processing of dynamic physical objects and spatial working memory. [[OSF](#)]

## CONFERENCE PRESENTATIONS

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**Maione, S. M., & Liu, S.** (2024). Frontoparietal regions engaged in physical prediction are also involved in spatial working memory. *Conference on Cognitive Computational Neuroscience*, Boston, MA. [[Conference abstract](#)]. [[Poster](#)].

Kim, M., **Maione, S. M.**, Drissi, A., Liu, S. (2024). Cortical regions preferentially engaged during social and physical processing represent obstacles to agent action and object motion. *Conference on Cognitive Computational Neuroscience*, Boston, MA. [[Conference abstract](#)].

**Maione, S. M.**, Gerhardstein, P. (2023). Perception through Visual Snow Syndrome: Faces, Scenes and Self-Reported Symptoms. *Binghamton University Psi Chi Fair*, Binghamton, NY. (**Undergraduate Research Award**).

**Maione, S. M.**, Duggan, N., Gerhardstein, P. (2022). Changing Orientation Anisotropy Influenced by the Digital Age. *Binghamton University Psi Chi Fair*, Binghamton, NY.

## SERVICE

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### Emerging Leaders Program Mentor

*Binghamton University*

*Supervisor: Tyler Lenga*

*May 2022 - May 2023*

- Provided a 90-minute, structured, weekly space for 13 newly matriculated students to develop professional skills and become acclimated to college life
- Led a partnership with the Binghamton Rescue Mission, a homeless shelter, and created care packages and presentations for residents

### Supplemental Instruction Leader

*Binghamton University*

*Program Supervisor: Lauren Morris*

*Aug 2021 - May 2023*

*Faculty Supervisor: Brittany Race*

- Created 60-minute, structured, weekly review sessions for an introductory psychology lecture, with attendance ranging from 20-135 students per session
- Connected students with campus resources and specific faculty to help students foster research experience and professional development

## HONORS/AWARDS

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**2023** Steven W. Kovacs Memorial Award

**2023** Binghamton University Undergraduate Research Award

**2022** Dr. Dominick A. and Susan G. Artuso Scholarships

## LABORATORY/TECHNICAL SKILLS

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*Laboratory:* Open science framework (pre-registration), fMRI (heudiconv, fMRIPrep, FSL, Nilearn), EEG (SSVEP, ERP), eye tracking (iView X), participant recruitment (Cloud research, MTURK, Prolific),

developmental populations (age 3 and above).

*Technical:* Python, R, machine learning (tidymodels, scikit-learn), MATLAB, PsychoPy, HPC computing (e.g. SLURM), Git, Jupyter, HTML/XML, Singularity, Docker, remote shell servers (SSH/SMB).

*Other:* Hardware installation & repair, patient consenting, Qualtrics, Adobe/Microsoft/Google/Apache Suite.